Phase 1 Preprocessing Strategy Report: An Explainable Risk-Triage System for Public Procurement Oversight in Kenya

Razan Ibrahim
Carnegie Mellon University Africa
roi@andrew.cmu.edu

Abstract

This report presents the data preprocessing strategy for a procurement risk-triage system built on Kenya's open contracting data. The dataset spans five linked CSV files covering 259,709 tender records, 109,163 awards, 79,870 supplier records, 109,163 contracts, and 813,078 party records published between July 2018 and August 2026. Because no ground-truth corruption labels exist, the task is formulated as unsupervised risk ranking, and the target is a composite risk score built from computable red-flag indicators. Every figure in this report was measured on the raw data. The audit shows that 60.6% of merged rows are tenders with no recorded award, so the scoring unit is the 109,163 awarded contracts. It also shows that the tender advertisement period cannot be measured reliably, so it is replaced by the decision period. The report diagnoses missingness mechanisms empirically and specifies justified strategies for cleaning, imputation, encoding, scaling, and temporal splitting.

1 GitHub Repository Architecture

The project repository is publicly accessible[1] and is organized according to the file hierarchy presented below

```
kenya-procurement-triage/
|-- .gitignore
|-- requirements.txt
|-- README.md
|-- data/
|   |-- raw/
|   `-- processed/
|-- phase1_strategy/
|   `-- Phase1_Strategy_Report.pdf
|-- phase2_implementation/
|   |-- notebooks/
|   |   `-- pipeline_execution.ipynb
|   `-- Phase2_Final_Report.pdf
|-- src/
|   |-- __init__.py
|   |-- transformers.py
|   `-- pipeline_builder.py
`-- figures/
```

The `.gitignore` excludes raw data, binaries, and virtual environments. Dependencies are pinned in `requirements.txt` to the versions used for the audit.

2 Dataset Overview and Problem Audit

2.1 Data Source

The dataset is downloaded from the Open Contracting Partnership Data Registry, Kenya PPRA[2] publication 147 [1]. It is distributed as a single compressed archive (`kenya_ppra_full.csv.tar.gz`, approximately 55 MB) containing five linked CSV files published to the Open Contracting Data Standard (OCDS)[3] [2]. The data is published as open government data under Executive Order No. 2 of 2018.

2.2 File Structure and Linking Keys

The five files and their row counts are summarized in Table 1. All files link back to `main.csv` via `main_ocid`. Awards link to suppliers via `awards_id` and to contracts via `awardID`.

File	Rows	Columns
main.csv	259,709	15
awards.csv	109,163	9
awards_suppliers.csv	79,870	5
contracts.csv	109,163	12
parties.csv	813,078	16
Merged working table	**277,325**	**23**
Scoring unit (awarded)	**109,163**	**23**

Table 1: File sizes, the merged working table ($N = 277{,}325$, $P = 23$), and the scoring unit.

[1] https://github.com/Razan-O-Elobeid/kenya-procurement-triage

[2] Public Procurement Regulatory Authority (PPRA): the Kenyan government body responsible for overseeing public procurement.

[3] Open Contracting Data Standard (OCDS): an international open-data specification for publishing procurement records.

The merged table is larger than `main.csv` because 91,547 tenders carry 109,163 awards between them. The remaining 168,162 tenders (60.6%) have no recorded award. Their award-stage fields do not exist rather than being missing, so they are audited but not scored. The scoring unit is the awarded contract, because that is the object an auditor reviews.

2.3 Machine Learning (ML) Task Type

This is an **unsupervised risk-ranking** task. No dataset anywhere labels a Kenyan contract as corrupt. Formulating this as supervised classification would require inventing labels, making any reported accuracy meaningless. The target Y is therefore a composite risk score built from red-flag indicators following the Corruption Risk Index (CRI) methodology [3, 4]. Each indicator f_k lies in $[0, 1]$. Following CRI practice, the score averages the indicators available for each contract:

$$R(c) = \frac{\sum_{k \in A(c)} w_k f_k(c)}{\sum_{k \in A(c)} w_k} \tag{1}$$

where $A(c)$ is the set of indicators computable for contract c. Equal weights are the default. The indicators are defined in Section 4.

2.4 Variable Inventory

Table 2 categorizes all 23 columns of the merged working table by data type.

2.5 Baseline Missingness Metrics

Table 3 records every column's missing count and percentage before any processing, at merged-table level and, for award-stage fields, among the 109,163 awarded rows. Mechanisms follow the standard classification: Missing Completely at Random (MCAR), Missing at Random (MAR, dependent on observed columns), and Missing Not at Random (MNAR, dependent on the unobserved value itself). A fourth category, *structural*, covers fields that do not exist because the tender never reached the award stage.

Across the merged matrix, 27.73% of cells are missing. Only one row is complete across all 23 columns, because `status` is almost always blank. Excluding `status` and `tender_awardCriteria`, which become flags, 38,538 rows (13.90%) are complete. Complete-case analysis is therefore not an option. The 105 recorded terminations almost certainly undercount reality, since contract failures are often not formally documented, so `is_terminated` is a valid but conservative signal.

2.6 Target and Feature Distributions

The target $R(c)$ cannot be profiled before its indicators are computed. There are no classes, so class imbalance does not apply. The rarest binary input is `is_terminated`, positive for 105 of 109,163 awards (0.10%).

`value_amount` has mean 15,409,360 KES, median 498,150 KES and maximum 160.6 billion KES. Its skewness is 161.0. After $\log(1 + x)$ the skewness is -1.38, and -0.51 once the 1,129 zero-value awards (1.03%) are set aside, so the zeros alone create most of the residual skew. The decision period (Section 4) has median 22 days and skewness -2.69, driven by contracts signed before bidding closed.

3 Cleaning and Imputation Strategy

3.1 Missingness Mechanism Treatment

Structural (award stage absent). The 168,162 unawarded tenders have no value, supplier or contract to review. They are not imputed. Scoring is restricted to awarded contracts, which removes the structural share of every award-stage column.

MNAR columns (missing because of what the value would have been). Imputing these would erase the signal, so each becomes a binary flag:

- `award_criteria_missing` = 1 if `tender_awardCriteria` is blank, else 0
- `supplier_missing` = 1 if an award has no named supplier (26.83% of awards), else 0
- `is_terminated` = 1 if status equals "terminated", else 0

MNAR cannot be proven from the data alone. The diagnosis rests on domain reasoning: an undocumented selection rule or an unnamed winner is itself a transparency failure.

MAR columns (missingness linked to an observed column). Missingness of `dateSigned` differs by procurement method (open 13.9%, selective 13.0%, direct 7.4%), which is evidence against MCAR. Where `dateSigned` is missing,

Column	Type	Role / Notes
ocid, id, tender_id, main_ocid	Nominal Categorical	Join keys only, drop before modeling
date	Datetime	Upload timestamp; records when a record was published, not used for splitting
tag	Nominal Categorical	Always "compiled", drop
language	Nominal Categorical	Always "en", drop
initiationType	Nominal Categorical	Always "tender", drop
buyer_id	Discrete	Buyer key for concentration and Benford features
buyer_name	Nominal Categorical	Buyer identity in plain text
tender_title	Text	Free-text description; not used in Phase 1
tender_mainProcurementCategory	Nominal Categorical	Three values: goods, works, services
tender_procurementMethod	Ordinal Categorical	Six values, grouped into three competition levels
tender_tenderPeriod_startDate	Datetime	When bidding opened; unreliable (Section 4)
tender_tenderPeriod_endDate	Datetime	Bidding deadline; temporal split key
tender_awardCriteria	Nominal Categorical	79.24% missing, converted to binary flag
value_amount	Continuous	Award value in KES (Kenyan Shilling)
value_currency	Nominal Categorical	Always "KES", drop
supplier_name	Text	Supplier identity; target of entity resolution
contractPeriod_startDate	Datetime	Contract start; feeds no indicator, display only
contractPeriod_endDate	Datetime	Contract end; feeds no indicator, display only
dateSigned	Datetime	Contract signing date; source of the decision period
status	Nominal Categorical	105 "terminated", otherwise blank; converted to flag

Table 2: Variable inventory for the merged working table ($P = 23$). `supplier_name` is classified as Text because its primary use is string matching for entity resolution, not categorical grouping.

the decision-period indicator f_3 is dropped from $A(c)$ in Equation 1 rather than imputed. For the isolation forest input `days_to_signing`, missing values are imputed with the median within each procurement method and recorded in a flag `signing_date_missing`. The median is used rather than the mean because the distribution is strongly skewed (-2.69).

The contract-period dates are also MAR, but they feed no indicator, so they are kept for display and excluded from model inputs. Imputing a field the model never uses adds no information.

`tender_mainProcurementCategory` (722 rows, 0.26%) was expected to be MCAR, but its missing rate is eleven times higher for selective tenders (1.78%) than for open tenders (0.16%). It is therefore treated as MAR and imputed with the mode within each procurement method. The 722 rows are spread over 256 buyers, with no buyer above 3.6%, so the gap is not one buyer's reporting habit.

The 248 records missing both tender-period dates cannot be placed in time. They are excluded at the split stage (Section 6.1) instead of being imputed. KNN and multiple imputation were rejected for these small gaps: they add computational cost for negligible gain.

3.2 Outlier Identification and Treatment

`value_amount` is heavily right-skewed (skewness 161.0), so the IQR rule is used for detection and the Z-score rule, which assumes a Gaussian distribution, is rejected:

$$\text{Lower} = Q_1 - 1.5 \times IQR$$
$$\text{Upper} = Q_3 + 1.5 \times IQR$$

With $Q_1 = 111{,}537$ and $Q_3 = 2{,}599{,}744$ KES, the upper bound is 6,332,053 KES and the lower bound is negative. 12.0% of awards exceed the upper bound. Capping at that bound would truncate one award in eight, mostly legitimate large works contracts.

Treatment: Winsorization at the 99th percentile (135,000,000 KES on the full data; the threshold is refit on the training set), not row deletion and not IQR capping. A row with an extreme value still carries valid signal in every other column, so deleting it loses information. Because extreme values matter in integrity analysis, a flag `above_p99` preserves the fact that a value was capped, and the raw value stays visible to auditors.

Zero-value awards (1,129, 1.03%) are not plausible for a signed award. Each is flagged in `zero_value_award` and excluded from the value-based indicators f_4 and f_5; a zero has no first digit for Benford's law. For the isolation forest value input, zeros are imputed with the me-

Column	Missing	% merged	% awarded	Mechanism and evidence
status	277,220	99.96	99.90	MNAR: only terminations are ever recorded
tender_awardCriteria	219,751	79.24	n/a	MNAR: selection criteria left undocumented
contractPeriod_startDate	205,933	74.26	34.60	Structural + MAR: varies by method, 28.8% to 40.6%
supplier_name	197,455	71.20	26.83	Structural + MNAR: winner not recorded
dateSigned	183,072	66.01	13.66	Structural + MAR: open 13.9%, selective 13.0%, direct 7.4%
contractPeriod_endDate	179,576	64.75	10.46	Structural + MAR: varies by method, 7.4% to 14.7%
value_amount, value_currency, main_ocid	168,162	60.64	0.00	Structural: tender has no award
tender_mainProcurementCategory	722	0.26	n/a	MAR: selective 1.78% vs open 0.16%; spread over 256 buyers
tenderPeriod start and end dates	248	0.09	n/a	Both missing together; record cannot be placed in time
remaining 12 columns	0	0.00	0.00	Complete

Table 3: Baseline missingness profile with measured evidence for each diagnosed mechanism. "n/a": tender-level field.

dian within procurement category, with the flag retained so the information is not lost.

3.3 Entity Resolution Strategy

Supplier identifiers in the awards data are not stable entity keys: 23.76% of normalized supplier names appear under more than one identifier, while no identifier carries more than one name. The identifier therefore marks a record, not a firm. Resolution has three stages:

1. **Exact:** normalize names (case, whitespace, punctuation, legal suffixes such as Ltd/Limited) and merge exact matches. This alone collapses the 23.76% of multi-identifier names.

2. **Fuzzy:** token sort ratio ≥ 90 within blocks sharing a first token, to catch spelling variants.

3. **Confirm:** `identifier_id` in `parties.csv`, present for 100% of supplier parties, confirms or vetoes each candidate merge.

The error rate is estimated on a manually labeled sample of 200 supplier-name pairs (precision and recall reported). High-risk suppliers are then checked against three external sources: the World Bank Debarred Firms list[4], the Kenya Gazette company register[5], and the OpenSanctions PEP[6] database.

4 Feature Engineering

Five indicators enter $R(c)$ in Equation 1, plus one candidate pending validation [4, 5, 6]. Table 4 records which indicators the release can and cannot support.

f_1: **is_direct_award.** Binary. 1 if `tender_procurementMethod` is "direct" or "SpeciallyPermitted", else 0. 8,800 tenders (3.4%) use these methods, which bypass competition.

f_2: **award_criteria_missing.** Binary. 1 if `tender_awardCriteria` is blank, else 0.

f_3: **decision period.** The days between the bidding deadline and contract signing. A very short decision period suggests the evaluation was decided in advance [3]:

$$\begin{aligned} d(c) &= t_{\text{signed}}(c) - t_{\text{close}}(c) \\ f_3(c) &= 1 - \hat{F}_{\text{train}}\big(d(c)\big) \end{aligned} \tag{2}$$

where $\hat{F}_{\text{train}}$ is the empirical distribution function of d on the training set, so shorter periods score higher and the score is bounded in $[0, 1]$.

[4] https://www.worldbank.org/en/projects-operations/procurement/debarred-firms

[5] https://new.kenyalaw.org/gazettes/

[6] Politically Exposed Persons (PEP): individuals who hold or have held prominent public positions. https://www.opensanctions.org/datasets/default/

$d(c)$ is available for 94,140 awards (86.2%). Its median is 22 days, and 7.6% of contracts were signed before bidding closed. These receive the highest scores: they are either irregular or recording errors, and both warrant review.

f_3 replaces the tender advertisement period originally planned. The audit found the advertisement period unusable: 42.2% of records have a start date after the end date, and among the rest the median duration is 0 days, with 92.8% under 7 days. This pattern reflects how dates were recorded, not how long tenders were open, so imputing it would manufacture a signal.

f_4: supplier_concentration. The share of buyer b's prior awarded value that went to supplier s, computed over an expanding window of awards dated before contract c, after entity resolution and excluding zero values:

$$f_4(c) = \frac{\sum_{c' \in C_{b,s}^{<t}} v(c')}{\sum_{c' \in C_b^{<t}} v(c')} \tag{3}$$

where $C_b^{<t}$ is the set of buyer b's awards before the date t of contract c, and $C_{b,s}^{<t}$ the subset won by s. It is unavailable for a buyer's first award. Values near 1 indicate dependence on a single supplier.

f_5: benford_deviation. Conformity of buyer b's prior non-zero award values to Benford's law, measured by mean absolute deviation (MAD) of first-digit proportions:

$$\begin{aligned} \mathrm{MAD}_b &= \frac{1}{9}\sum_{d=1}^{9} \left|\hat{p}_{b,d} - E_d\right| \\ E_d &= \log_{10}\left(1 + \tfrac{1}{d}\right) \end{aligned} \tag{4}$$

and $f_5(c) = \hat{F}_{\text{train}}(\mathrm{MAD}_b)$. MAD is chosen over the chi-squared statistic because chi-squared grows with sample size, so large buyers would score higher for the same proportional deviation. f_5 requires at least 100 prior awards; below that it is unavailable.

f_6 (candidate): single_supplier_party. The release has no tenderer count, so single bidding, the most validated CRI indicator [4], cannot be read directly. However, 79.3% of processes list more supplier parties than award winners (median 3), consistent with bidders being recorded as parties. $f_6 = 1$ if a process lists exactly one supplier party. It enters $R(c)$ only if a manual check of 50 processes against tender notices on the PPRA portal confirms that extra supplier parties are bidders.

5 Encoding, Scaling, and Transformation Strategy

5.1 Categorical Encoding

Ordinal encoding: `tender_procurementMethod`. The six observed methods are grouped into three competition levels: open = 0 (233,594); restricted = 1 (selective 17,268, AlternativeSelection 41, community participation 6); non-competitive = 2 (direct 8,744, SpeciallyPermitted 56). The rare methods are too small to support levels of their own, and there is no defensible order between direct and specially permitted awards, since both bypass open competition. One-hot encoding was rejected because it would discard the ordering of competition levels.

One-hot encoding: `tender_mainProcurementCategory`. Goods, works and services have no natural order, so ordinal encoding was rejected. One level is dropped to avoid perfect collinearity in the VIF screen and PCA; the isolation forest is unaffected either way.

Binary flags: `award_criteria_missing`, `supplier_missing`, `is_terminated`, `zero_value_award`, `above_p99`, `signing_date_missing`.

Dropped: `language`, `initiationType`, `tag` and `value_currency` (constant), identifier columns used only for joining, and the contract-period dates (no indicator uses them).

5.2 Numerical Scaling

`value_amount`: Winsorize, log, then standardize. The isolation forest [7] chooses a split value uniformly between a feature's minimum and maximum. On the raw scale, with a median of 498,150 KES and a maximum of 160.6 billion KES, almost every split would fall in the empty upper range, so the log transform genuinely changes model behavior:

$$x_v(c) = \log\left(1 + \min(v(c), q_{0.99}^{\text{train}})\right) \tag{5}$$

$\log(1+x)$ is used rather than $\log(x)$ because $\log(0)$ is undefined. StandardScaler ($\mu = 0$,

Indicator	Source fields	Coverage (awarded contracts)	Status
f_1 direct award	procurementMethod	100%	Used
f_2 criteria missing	awardCriteria	100%	Used
f_3 decision period	dateSigned, tender end date	86.2%	Used
f_4 supplier concentration	value, resolved supplier	all but a buyer's first award	Used
f_5 Benford MAD	value	buyers with ≥ 100 prior awards	Used
f_6 single supplier party	parties	79.3% list extra parties	Pending validation
Advertisement period	tender start and end dates	42.2% negative	Rejected
Single bidding (tenderer count)	tender.tenderers	absent from release	Not computable

Table 4: Indicator coverage map: which integrity indicators the Kenyan release can and cannot support.

$\sigma = 1$) is then applied. It does not affect the isolation forest, which is invariant to per-feature affine transforms, but it is required for PCA and the VIF screen, where unequal variances would let one feature dominate. MinMaxScaler was rejected because it is sensitive to the remaining extreme values and would compress most observations near zero.

`days_to_signing`: the decision period $d(c)$ includes negative values (7.6%), so $\log(1 + x)$ cannot be applied. A signed log transform $\text{sign}(d)\log(1 + |d|)$ compresses both tails, followed by StandardScaler for the same reasons.

Indicators f_1 to f_6 lie in $[0, 1]$ by construction, either binary or empirical-CDF scores, so they need no further scaling for $R(c)$. They are standardized together with other inputs only for PCA.

5.3 Feature-to-Target Statistical Test Matching Matrix

Because $R(c)$ is built from the indicators, correlating an indicator with $R(c)$ would be circular: the association is guaranteed by construction. Each indicator f_k is therefore tested against the leave-one-out score $R_{-k}(c)$, which excludes f_k. Inputs that are not components are tested against $R(c)$. The procurement method and `days_to_signing` are tested against R_{-1} and R_{-3}, since f_1 and f_3 are functions of them. Table 5 gives the chosen test, the rejected alternative, and the reason.

5.4 Multicollinearity Protocol

Pairwise Spearman $|\rho| \geq 0.85$ is applied to continuous inputs; Spearman is used because the inputs are skewed and bounded. The Variance Inflation Factor (VIF) > 10 on standardized inputs is a secondary check. When two features are collinear, the one with higher coverage is kept, and on a tie the literature-validated indicator is kept. Association with $R(c)$ is never used as the tie-break, because that would reintroduce circularity.

Two pairs are collinear by construction and are resolved in advance. f_3 is a monotone transform of `days_to_signing`, and f_1 is the top level of the method encoding. The isolation forest receives the richer variable of each pair (`days_to_signing` and the method level), while $R(c)$ uses the indicators. Any further pairs are caught empirically in Phase 2 with the correlation heatmap and collinear-pair scatterplots.

6 Split Strategy and Leakage Prevention

6.1 Temporal Split

The dataset spans July 2018 to August 2026. A random split would let a model trained on 2025 contracts score 2024 contracts, which is look-ahead bias. The data is split by **tender end date**. Upload dates were rejected because they record when a record was published, not when the procurement happened: tenders closing in 2021 carry 2025 upload timestamps.

Among awarded contracts, records from 2018 to 2023 form the training set (72,585, 66.6%) and records from 2024 to 2026 the test set (36,399, 33.4%). The 179 awards (0.16%) with a missing or out-of-window end date are excluded. This mirrors deployment: a model trained on history scores contracts as they arrive.

Inside the training period, `TimeSeriesSplit` with 5 folds is used for hyperparameter selection, each validation fold strictly following its training fold. With no labels, the selection criterion is rank stability: the Spearman correlation between the score rankings of the same validation fold under different random seeds. The number of trees and the subsample size are chosen to maximize it. The contamination parameter is not tuned, because the output is a ranking, not a binary label.

Feature	Type	Target	Chosen / Rejected / Reason
f_1 is_direct_award	Binary	R_{-1}	**Chosen:** point-biserial r. **Rejected:** chi-square, which needs a categorical target.
f_2 award_criteria_missing	Binary	R_{-2}	Same as f_1.
f_3 decision-period score	Continuous, $[0, 1]$	R_{-3}	**Chosen:** Spearman ρ. **Rejected:** Pearson r. R mixes binary and rank-based components, so it is not Gaussian, and d has skewness -2.69.
f_4 supplier_concentration	Continuous, $[0, 1]$	R_{-4}	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Shares typically pile up at 0 and 1; normality is not assumed.
f_5 benford_deviation	Continuous, $[0, 1]$	R_{-5}	Same as f_4.
f_6 single_supplier_party	Binary	R_{-6}	Same as f_1, if validated.
value_amount (log)	Continuous	R	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Skewness -0.51 excluding zeros and -1.38 with them.
days_to_signing	Continuous	R_{-3}	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Skewness -2.69.
tender_procurementMethod	Ordinal, 3 levels	R_{-1}	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Codes 0–2 are ranks; the intervals between them are not equal.
tender_mainProcurement-Category	Nominal, 3 levels	R	**Chosen:** Kruskal-Wallis H with ε^2 effect size, plus mutual information. **Rejected:** ANOVA F, which assumes normal within-group distributions that a bounded, non-Gaussian R violates; Pearson and Spearman, which need an ordering.
supplier_missing, zero_value_award	Binary	R	**Chosen:** point-biserial r, as for f_1.
is_terminated	Binary, 0.10% positive	R	**Chosen:** Mann-Whitney U with rank-biserial effect size. **Rejected:** point-biserial r, which is unstable when only 105 of 109,163 cases are positive.

Table 5: Feature-to-target statistical test matching matrix. Indicators are tested against leave-one-out scores to avoid circularity.

6.2 Entity-Level Leakage

Supplier concentration and Benford deviation depend on a buyer's history, so a buyer appearing in both sets is the main leakage risk. Both are computed over an expanding window: a contract's buyer-level features use only that buyer's awards dated before it. The same rule applies to training, test and deployment, so no test information reaches training. GroupKFold was rejected: separating a buyer's history from its future is exactly what these features need, and here the leakage risk is temporal ordering, not shared identity.

6.3 Transformer Fit Architecture

All transformers (imputers, Winsorization thresholds, scalers, encoders and the empirical distribution functions behind f_3 and f_5) are fit strictly on `X_train`, as shown in Algorithm 1. The `sklearn.pipeline.Pipeline` and `ColumnTransformer` wrappers enforce this by design.

6.4 Target Leakage Prevention

$R(c)$ is computed only from the contract's own record and from history dated before it, with every distribution function fit on training data. No future information enters any feature. Table 6 summarizes all planned decisions for the Phase 2 decision log.

Algorithm 1: Leakage-free preprocessing and scoring pipeline using Isolation Forest [7], implemented via `sklearn.pipeline.Pipeline` and `ColumnTransformer`.

Input: X_train, X_test: awarded contracts ordered by tender end date
Output: Fitted transformers; $R(c)$ and isolation forest scores for X_test

Fit on X_train only:
Impute category (mode within method) and days_to_signing (median within method), with flags
Winsorize value at the training 99th percentile, then apply $\log(1 + x)$; signed log for days_to_signing
Fit OrdinalEncoder (3 competition levels), OneHotEncoder (drop first), StandardScaler
Fit empirical distribution functions for f_3 and f_5
Fit IsolationForest on the engineered inputs

For every contract, in date order:
Compute f_4 and f_5 from the buyer's awards dated before it

Transform X_test (no refitting):
Apply all fitted transformers; score $R(c)$ and IsolationForest.decision_function

Preprocessing Decision	Justification	Method
Unawarded tenders	60.6% have no award stage	Audited, not scored
MNAR missingness	Missingness is itself a risk signal	Binary flag
dateSigned (MAR)	Rate varies by method (7.4% to 13.9%)	f_3 unavailable; median within method + flag
Category (MAR)	Selective 1.78% vs open 0.16%	Mode within method
Contract-period dates	Feed no indicator	Excluded from model inputs
Missing tender dates	Cannot be placed in time	Excluded at split
Zero-value awards	1,129 implausible values	Flag; excluded from f_4, f_5
Extreme values	12.0% exceed IQR bound; skewness 161.0	Winsorize at training P99 + flag
value_amount transform	Uniform splits on a raw scale fail	$\log(1 + x)$, then StandardScaler
days_to_signing transform	7.6% negative, skewness -2.69	Signed log, then StandardScaler
Advertisement period	42.2% negative, median 0 days	Rejected; replaced by decision period
Benford measure	Chi-squared grows with sample size	MAD, ≥ 100 prior awards
tender_procurementMethod	Ordered competition levels	Ordinal encoding 0–2
tender_mainProcurementCategory	No natural order	One-hot encoding
Constant columns	Zero variance	Drop
Feature-target tests	Indicators are components of R	Leave-one-out score R_{-k}
Train/test split	Prevent look-ahead bias	Tender end date, 66.6/33.4
Hyperparameter search	No labels available	TimeSeriesSplit, rank stability
Buyer-history leakage	Features need buyer history	Expanding window
Transformer fit	Must not see test data	Fit on X_train only via Pipeline

Table 6: Pre-implementation decision log for the Phase 2 strategy-vs-reality comparison.

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